

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – ANALYTICAL PROBLEM SOLVING

Course Code:	CSA06	Course Title:	Design and Analysis of Algorithms
CO Assessed:	CO2 – Manipulation (BL1, BL2, BL3)	Assessment:	Assessment Tool 1 – Analytical Problem Solving
Weightage:	40% of CIA	Total Marks:	100
CO2 Statement:	Apply Brute Force technique to solve sorting, searching, Closest-Pair and Convex-Hull problems (BL1, BL2, BL3)		
Student Name:	Manoj M	Reg. No.:	192511060
Section / Batch:	CSE	Date:	27.07.26

Instructions to Students

- Answer the following question. Total Marks: 100.
- Analyze each problem carefully before applying the appropriate Brute Force technique.
- Clearly show all intermediate steps, comparisons, iterations, and logical reasoning used in solving the problems.
- Apply suitable algorithms for sorting, searching, Closest-Pair, and Convex-Hull problems wherever required.
- Justify the correctness and efficiency of the proposed solution using appropriate complexity analysis.
- Use diagrams, tables, or stepwise illustrations wherever necessary for better explanation.
- Maintain neat presentation, proper algorithmic notation, and structured analytical reasoning throughout the answers.

Question Type	Focus Area	Questions
Application-Based Questions	Apply Brute Force techniques for sorting, searching, Closest-Pair and Convex-Hull problem analysis	Q1

SECTION A – APPLICATION BASED QUESTIONS

Q51. Selection Sort – Random Integer Arrangement

Apply the Brute Force technique using Selection Sort on the dataset [38, 12, 47, 5, 29, 18]. Clearly interpret the problem and justify why Selection Sort is suitable for this unsorted dataset. Perform the sorting step-by-step and show the intermediate array after each pass. Count the total number of comparisons and swaps made during execution. Analyze the time complexity in best, average, and worst cases. Verify the correctness of the final sorted output and interpret the efficiency of the algorithm

Code of Conduct

I Manoj M(192511060) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

RUBRICS FOR CALCULATION

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Problem Context	20%	Clearly understands the problem and accurately identifies all requirements	Good understanding with minor gaps	Basic understanding; some requirements unclear	Poor understanding or misinterpretation of the problem
Application of Concepts	30%	Accurately applies appropriate concepts to solve complex problems	Applies relevant concepts correctly with minor errors	Applies basic concepts with limited accuracy	Fails to apply appropriate concepts
Problem-Solving Approach	20%	Logical, well-structured, and efficient approach	Mostly logical approach with minor gaps	Basic approach with some inconsistencies	Illogical or unclear approach
Accuracy of Solution	20%	Completely correct solution with proper justification	Mostly correct with minor errors	Partially correct solution	Incorrect or incomplete solution
Clarity	10%	Clear, well-organized, and properly explained solution	Understandable with minor clarity issues	Limited clarity and explanation	Poorly presented and difficult to understand

Marks Evaluation:

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Problem Context	20%				
Application of Concepts	30%				
Problem-Solving Approach	20%				
Accuracy of Solution	20%				
Clarity	10%				
Total Marks (100):					

Faculty In-charge	Course Coordinator	Head of Department
Name:	Name:	Name:
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

Selection Sort – Random Integer Arrangement

1. Problem Interpretation

Problem solving (Analytics)

1. Problem interpretation:

* The given data is $[12, 42, 18, 29]$ is an unsorted array of integers. The objective is to arrange the elements in ascending order.

2. Justification for using selection sort:

* selection sort is suitable for this dataset because it is a simple comparison-based sorting algorithm that efficiently sorts.

* It repeatedly selects the smallest element from the unsorted portion of the array and places it in its correct position.

3. working principle:

* selection sort divides the array into a sorted and an unsorted portion. During each pass.

* It finds the smallest element in the unsorted portion and swaps it with the first unsorted element.

2. Scenario-Based – Execution

Initial Array

[38, 12, 47, 5, 29, 18]

Pass 1

Find the smallest element from index 0 to 5.

38 12 47 5 29 18

↑

Minimum = 5

Swap 38 and 5.

[5, 12, 47, 38, 29, 18]

Comparisons = 5

Swaps = 1

Pass 2

Find the smallest element from index 1 to 5.

12 47 38 29 18

↑

Minimum = 12

Already in correct position.

[5, 12, 47, 38, 29, 18]

Comparisons = 4

Swaps = 0

Pass 3

Find the smallest element from index 2 to 5.

47 38 29 18

↑

Minimum = 18

Swap 47 and 18.

[5, 12, 18, 38, 29, 47]

Comparisons = 3

Swaps = 1

Pass 4

Find the smallest element from index 3 to 5.

38 29 47

↑

Minimum = 29

Swap 38 and 29.

[5, 12, 18, 29, 38, 47]

Comparisons = 2

Swaps = 1

Pass 5

Find the smallest element from index 4 to 5.

38 47

↑

Minimum = 38

Already in correct position.

[5, 12, 18, 29, 38, 47]

Comparisons = 1

Swaps = 0

Intermediate Arrays

Pass Array

Initial [38, 12, 47, 5, 29, 18]

Pass 1 [5, 12, 47, 38, 29, 18]

Pass Array

Pass 2 [5, 12, 47, 38, 29, 18]

Pass 3 [5, 12, 18, 38, 29, 47]

Pass 4 [5, 12, 18, 29, 38, 47]

Pass 5 [5, 12, 18, 29, 38, 47]

Total Comparisons

Selection Sort always performs

$(n-1)+(n-2)+\dots+1$

For $n = 6$

$5 + 4 + 3 + 2 + 1 = 15$

Total Comparisons = 15

Total Swaps

Actual swaps performed:

- **Pass 1 ✓**
- **Pass 2 ✗**
- **Pass 3 ✓**
- **Pass 4 ✓**

- Pass 5 ✗

Total Swaps = 3

3. Debugging & Optimisation Task – Execution Time Complexity Analysis

Case	Time Complexity	Reason
Best Case	$O(n^2)$	Comparisons remain the same even if already sorted.
Average Case	$O(n^2)$	Every pass scans the remaining unsorted elements.
Worst Case	$O(n^2)$	Same number of comparisons as all other cases.
Space Complexity	$O(1)$	Only one temporary variable is used during swapping.

RUBRICS FOR CALCULATION

Analytical Problem Solving					
Criteria	Weight age	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding ; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations

Interpretati on of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretatio n with minor omissions	Limited or unclear interpretation	No interpretatio n
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